Lab 5: ARMA estimation, diagnostics and forecasting

Reference solution

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Reference Python code in `correction.py`*. All numerical values below are obtained with* `numpy.random.seed(42)`.

Exercise 1 – AR(2) simulation and estimation

Solution

The PACF of the simulated series cuts off after lag 2, suggesting AR(2). Fit comparison:

order	log-lik.	AIC	BIC	LB(24) p
$(1, 0, 0)$	-720.28	1446.55	1459.19	0.0000
$(2, 0, 0)$	-684.91	**1377.81**	**1394.67**	0.7506
$(3, 0, 0)$	-684.84	1379.68	1400.76	0.7498

The MLE on $(2, 0, 0)$ gives $\widehat{\phi}_1 = +0.7051$, $\widehat{\phi}_2 = -0.3645$, close to the true $(0.6, -0.3)$ given $T = 500$. AR(1) is rejected by both AIC and Ljung–Box; AR(3) has a marginally larger BIC — AR(2) is the parsimonious choice.

Exercise 2 – MA(1) identification

Solution

Empirical $\widehat{\rho}(h)$ for $h = 1, \ldots, 5$:

$$(0.517,\ 0.024,\ -0.059,\ -0.073,\ -0.069),$$

the ACF cuts off at lag 1. The theoretical $\rho(1) = 0.7/1.49 = 0.4698$ is recovered to within ± 0.05.
Fit comparison:

order	log-lik.	AIC	BIC	LB(24) p
$(0, 0, 1)$	-688.01	**1382.02**	**1394.67**	0.6726
$(1, 0, 0)$	-732.61	1471.22	1483.86	0.0000

$\widehat{\theta} = +0.7558$ (true 0.7). The AR(1) alternative fails Ljung–Box because an MA(1) has only one non-zero autocorrelation, while AR(∞) is needed to reproduce that single spike via an autoregressive model: an AR(1) approximation leaves persistence on every lag, hence the residual autocorrelation.

Exercise 3 – ARMA$(1,1)$ on simulated data

Solution

order	log-lik.	AIC	BIC	LB(24) p
$(1,0,0)$	-720.66	1447.31	1459.96	0.0000
$(0,0,1)$	-728.47	1462.93	1475.58	0.0000
$(1,0,1)$	-687.45	1382.91	**1399.77**	0.6568
$(2,0,2)$	-684.82	**1381.64**	1406.93	0.8687

AIC marginally favours $(2,0,2)$, BIC favours $(1,0,1)$. The $(2,0,2)$ model adds three extra parameters for a 1.27-point gain in log-likelihood — classic over-fitting (AR/MA cancellation between the spurious factors). The parsimonious choice is $(1,0,1)$ with $\widehat{\phi} = +0.5093$ and $\widehat{\theta} = +0.4730$, very close to the true $(0.5, 0.4)$.

Exercise 4 – Box–Jenkins on the Atlanta residual

Solution

On the Atlanta decomposition residual ($n = 1644$, sd $= 2.884$):

order	log-lik.	AIC	BIC	$\widehat{\sigma}^2$	LB(24) p
$(1,0,0)$	-4071.62	8149.24	8165.46	8.293	0.0000
$(2,0,0)$	-4065.35	8138.70	8160.32	8.230	0.0000
$(0,0,1)$	-4071.22	8148.44	8164.65	8.289	0.0000
$(1,0,1)$	-3928.70	7865.39	7887.01	6.945	0.0000
$(2,0,1)$	-3871.89	**7753.78**	**7780.81**	6.480	0.0000

ARMA$(2,1)$ wins both AIC and BIC. The Ljung–Box test still rejects, however: the residual carries structure beyond what a linear ARMA can capture (residual seasonality / volatility clusters — to be addressed in Chapter 6).
Best parsimonious choice: **ARMA**$(2,1)$ as a useful approximation, with the caveat that the model is not a perfect fit.

Exercise 5 – Forecasting

Solution

With ARMA$(2,1)$, the 24-step forecast (figure `lab5_ex5_forecast.png`) returns smoothly to the unconditional mean of zero, and the 95% confidence band widens toward

$$\pm 1.96\,\widehat{\sigma}\sqrt{\sum_{j=0}^{\infty}\psi_j^2} \;=\; \pm 1.96 \cdot (\text{unconditional sd of } X_t).$$

Hold-out validation. Splitting at 80% ($n_{\text{train}} = 1315$, $n_{\text{test}} = 329$): out-of-sample RMSE $= 2.699$, to be compared with the in-sample residual sd $= 2.598$. The out-of-sample error is only $\approx 4\%$ larger than the in-sample sd: the model generalises reasonably well, but the seasonal / volatility structure noted in Exercise 4 shows up as small systematic forecast errors.